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INTRODUCTION TO

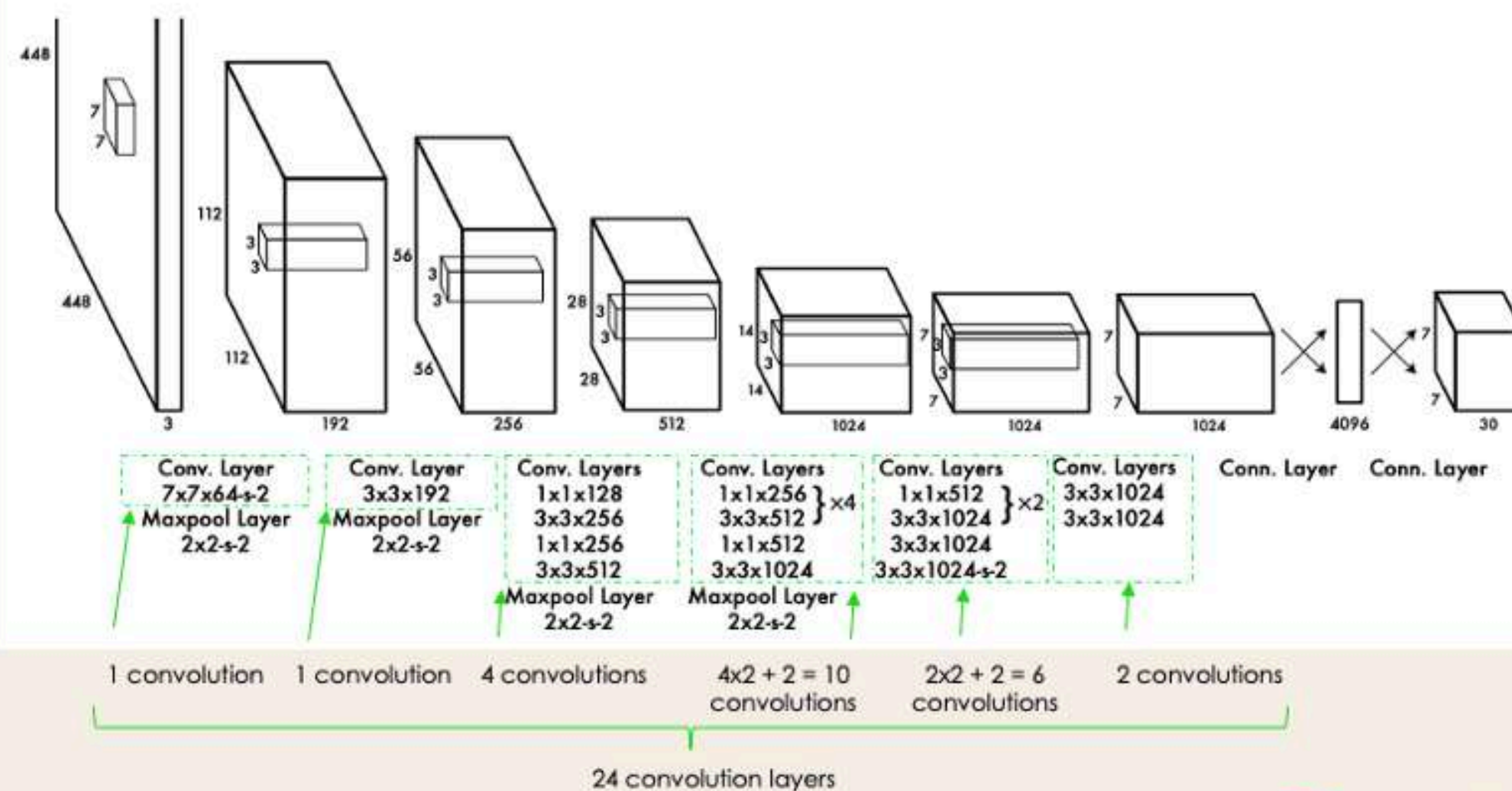
YOLO



27 December, 2025

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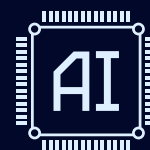
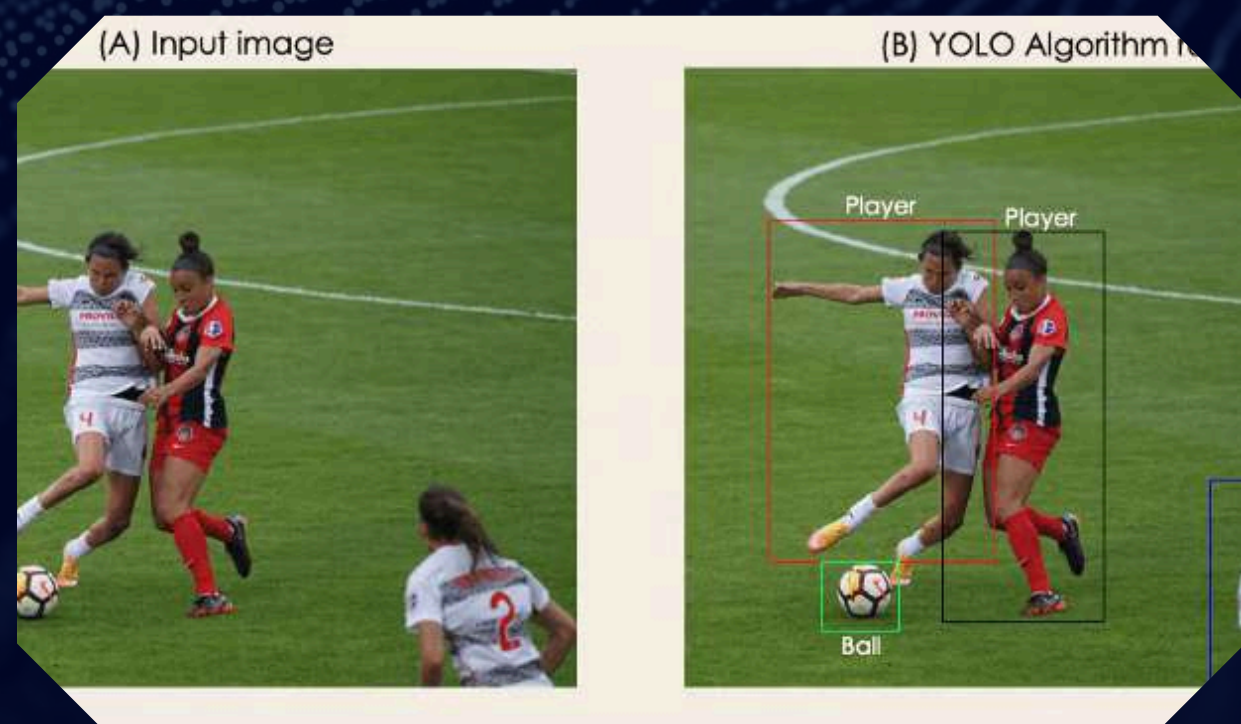
ARCHITECTURE



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HOW DOES YOLO OBJECT DETECTION WORK

UNDERSTANDING THE DISTINCTIONS



Residual blocks

This first step starts by dividing the original image (A) into $N \times N$ grid cells of equal shape, where N , in our case, is 4, as shown in the image on the right. Each cell in the grid is responsible for localizing and predicting the class of the object that it covers, along with the probability/confidence value.



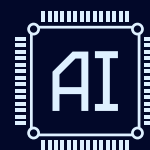
Bounding box regression

The next step is to determine the bounding boxes corresponding to rectangles, highlighting all the objects in the image. We can have as many bounding boxes as there are objects within a given image.



Intersection Over Unions or IOU

Most of the time, a single object in an image can have multiple grid box candidates for prediction, even though not all are relevant. The goal of the IOU (a value between 0 and 1) is to discard such grid boxes to only keep those that are relevant.



Non-Max Suppression or NMS

Setting a threshold for the IOU is not always enough because an object can have multiple boxes with IOU beyond the threshold, and leaving all those boxes might include noise. Here is where we can use NMS to keep only the boxes with the highest probability detection score.

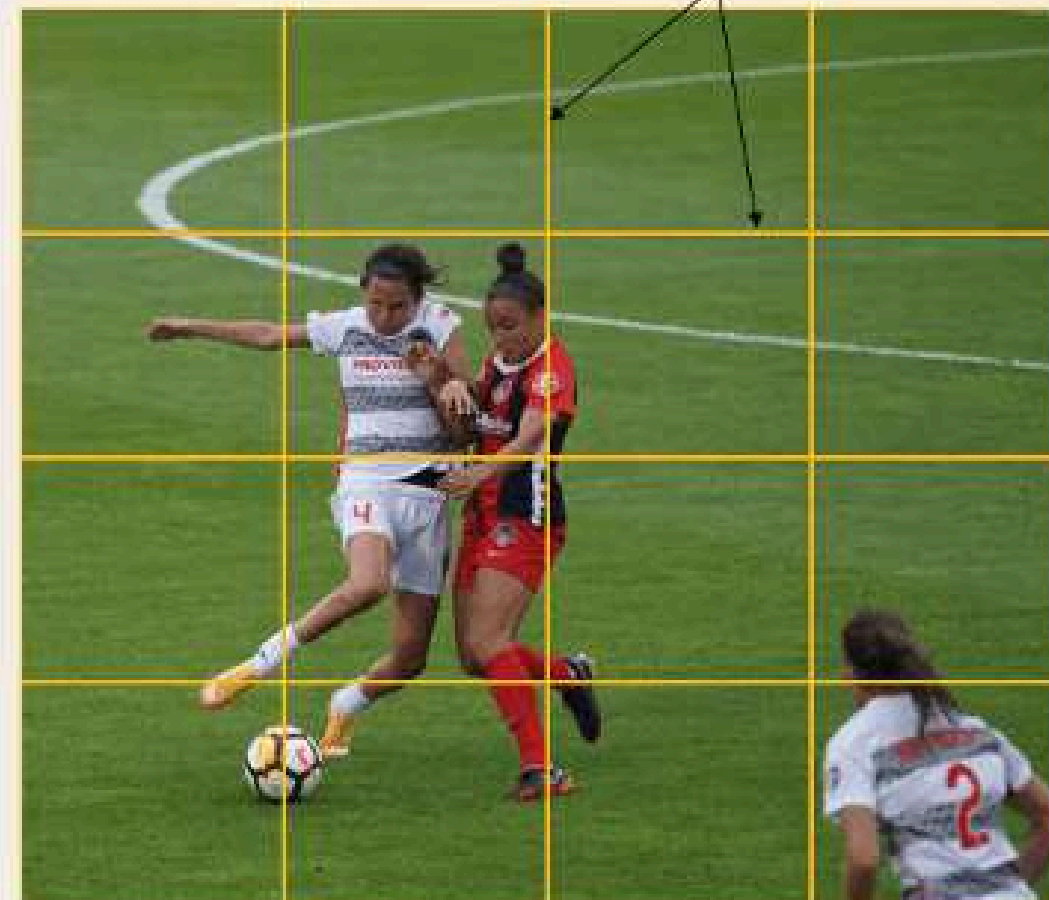
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RESIDUAL BLOCKS

Original input Image



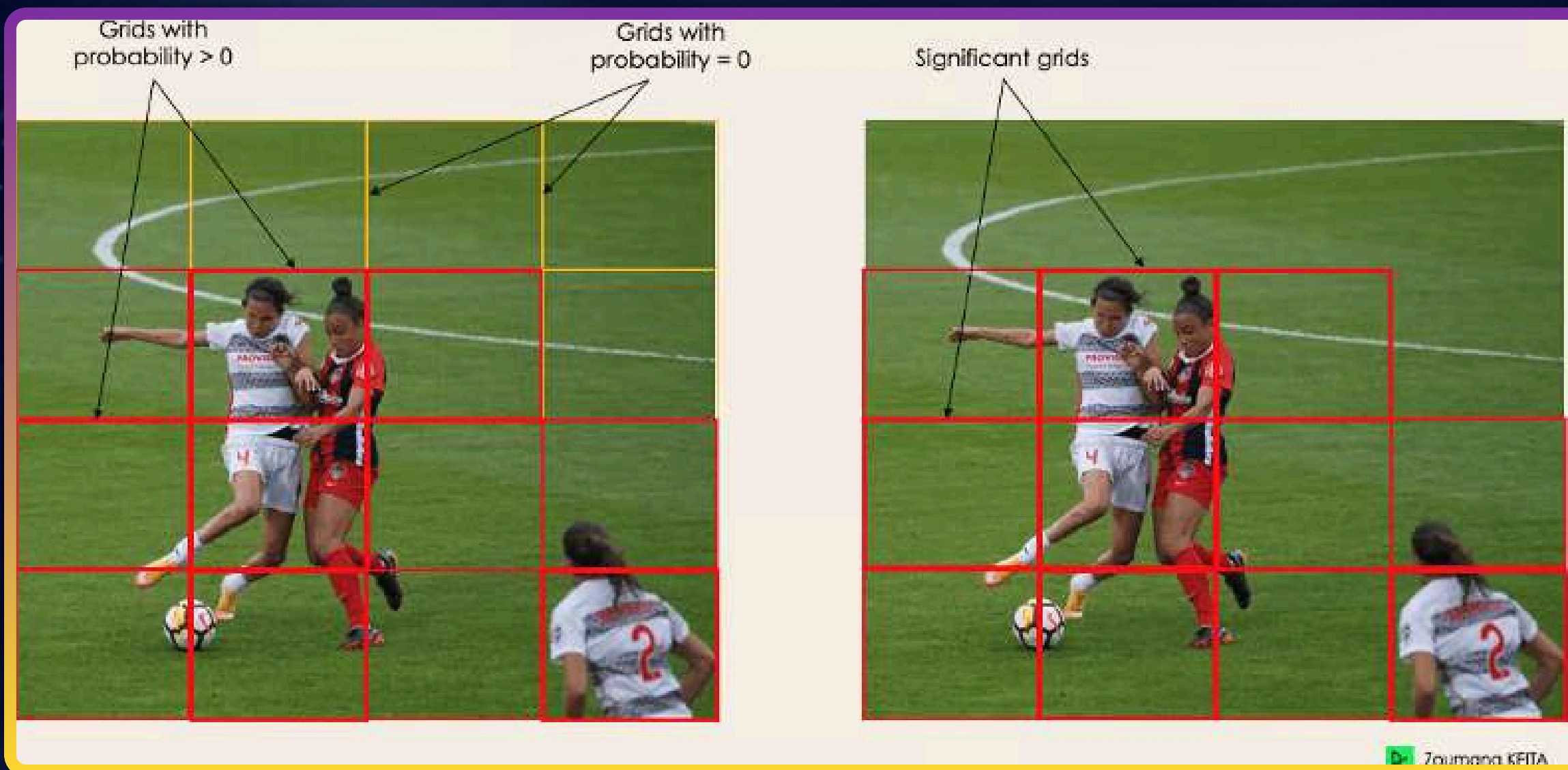
4x4 grid cells



 Zoomania KETA

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BOUNDING BOX REGRESSION



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INTERSECTION OVER UNIONS OR IOU

Diagram illustrating IOU calculation with two grids. Grid 1 has a low IOU ($\text{IOU Grid 1} < 0.5$) and Grid 2 has a high IOU ($\text{IOU Grid 2} > 0.5$). Grid 2 is selected.

IOU = $\frac{\text{Intersection Area}}{\text{Union Area}}$

Grid 2 selected

IOU visualization

$$\text{IOU} = \frac{\text{area of overlap}}{\text{area of union}} = \frac{\text{area of overlap}}{\text{area of union}}$$

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NON-MAX SUPPRESSION OR NMS

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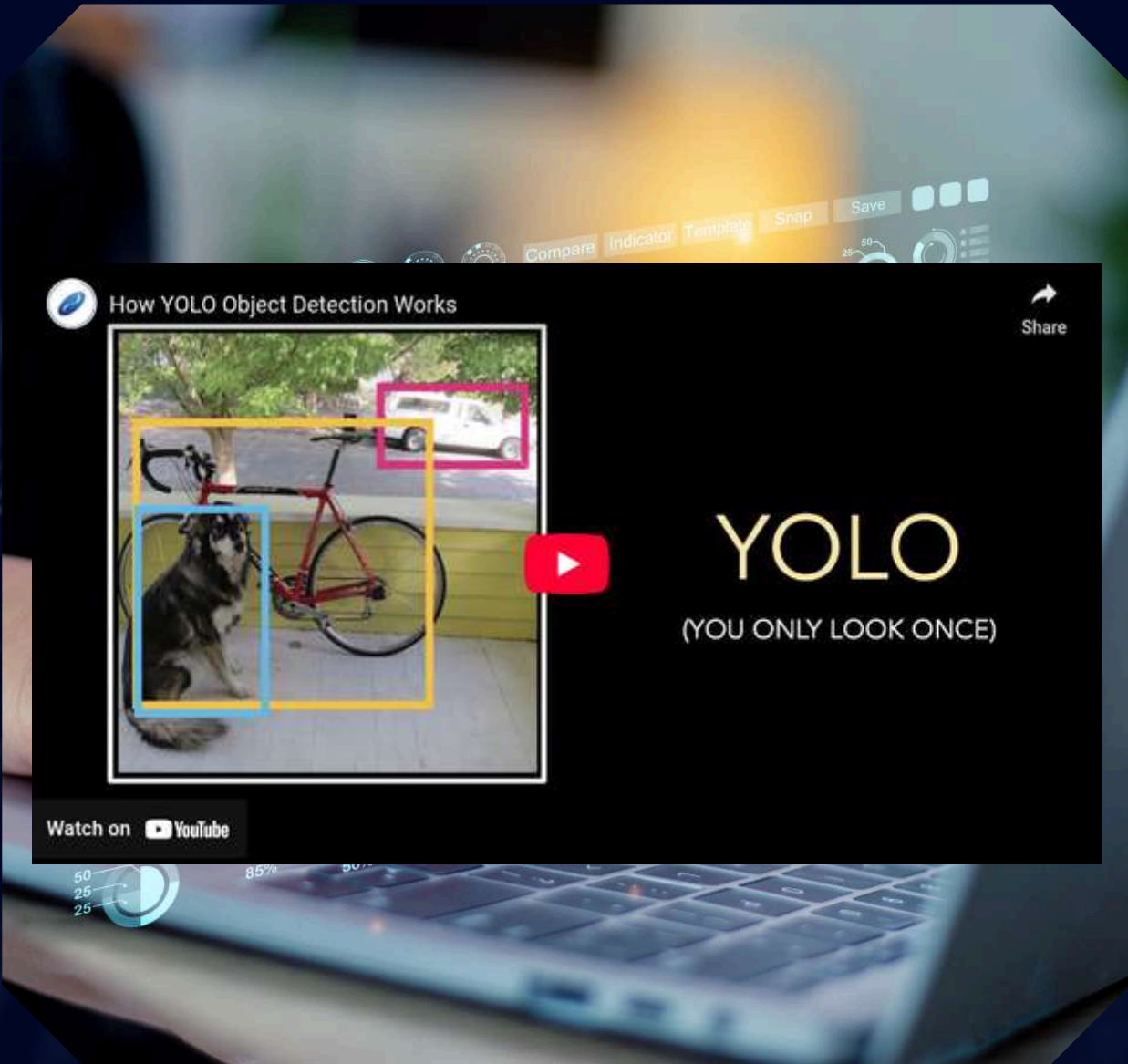
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OVERALL WORKFLOW



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NEURAL NETWORK ARCHITECTURES

Neural network architectures define how deep learning models process and analyze data. Different architectures specialize in specific tasks, optimizing performance for various applications.

NEURAL NETWORK ARCHITECTURES

Convolutional Neural Networks (CNNs) are widely used in image processing and computer vision. They employ convolutional layers to extract features from images, reducing the number of parameters while preserving essential details. Recurrent Neural Networks (RNNs) are designed for sequential data, such as speech recognition and language modeling, utilizing memory units to retain past information for better predictions.



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STEP TO DO



Load Dataset

Data Quality Matters: High-quality, labeled data is crucial for training accurate deep learning models. Poor data can introduce biases, leading to inaccurate predictions and unreliable outcomes.

Load YOLO & Train & Predict

Deep learning thrives on vast datasets. The more data available, the better the model's performance. Companies invest in scalable infrastructure to handle large-scale data processing.



Compute the yaml data file

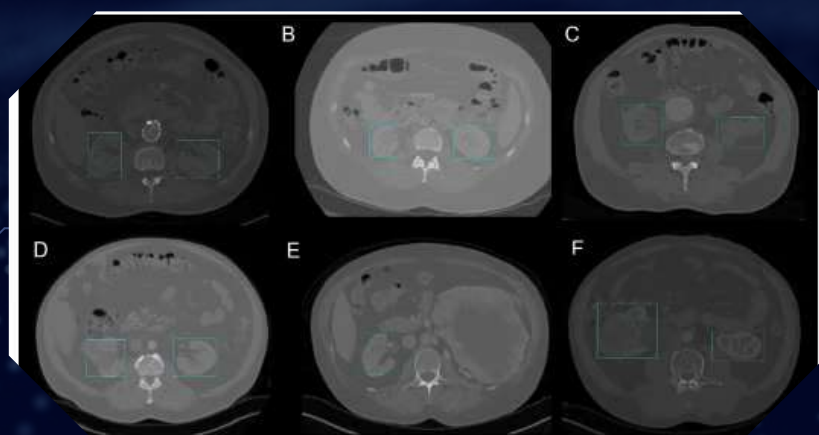
Cleaning and augmenting data help improve model accuracy. Techniques like normalization, augmentation, and feature scaling enhance the quality of input data, ensuring better generalization of deep learning models.

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SOME APPLICATION

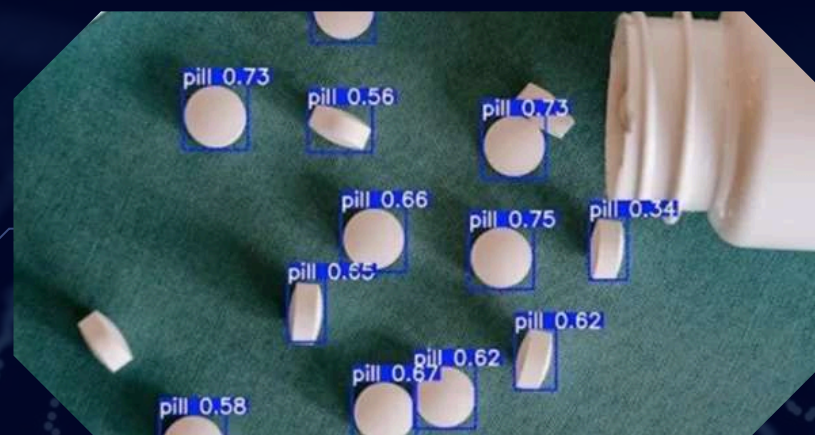


LIMITATIONS AND OBSTACLES



Health Care

Training deep learning models requires immense computational resources, including GPUs and TPUs, making it expensive for many organizations.



Counting

Handling large datasets raises privacy issues, requiring strict regulations and secure storage to protect user information.



Agriculture

Deep learning models often function as black boxes, making it difficult to interpret decisions. Addressing bias and ensuring fairness remains a major challenge in AI ethics.

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ASSIGNMENT

REAL-WORLD
SUCCESS
STORIES



Face Mask Dataset (YOLO Format)

<https://www.kaggle.com/datasets/aditya276/face-mask-dataset-yolo-format>



Construction Site Safety Image Dataset Roboflow

<https://www.kaggle.com/datasets/snehilsanyal/construction-site-safety-image-dataset-roboflow>



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